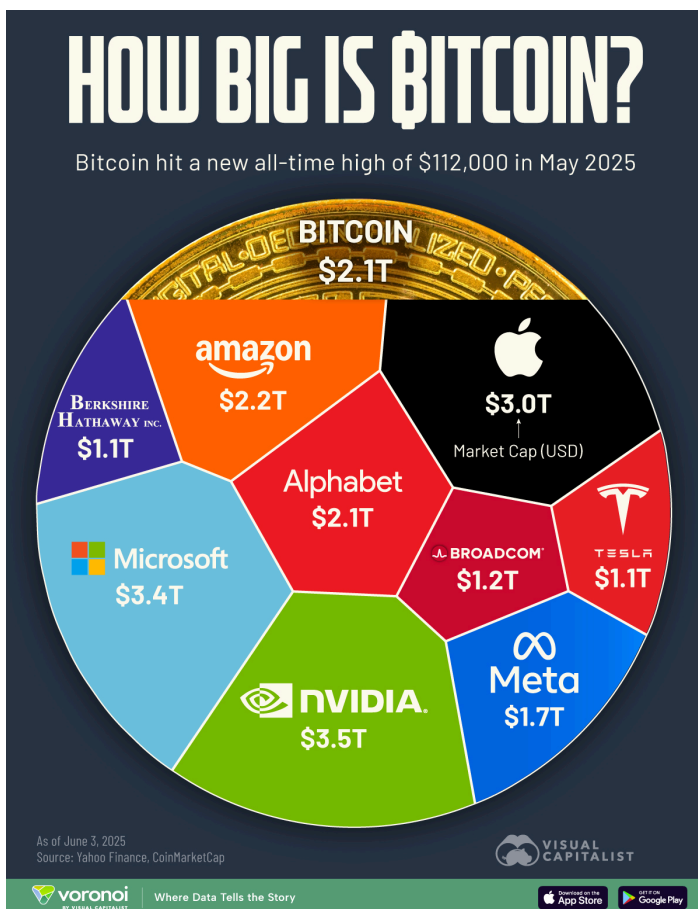


Quantitative Analysis and Valuation of *iShares Bitcoin Trust (IBIT)* Call Options

20th March 2026



Overview

The integration of Bitcoin into **traditional financial markets** through securitization in **Exchange-Traded Funds (ETFs)** (January 2024) has marked a turning point for the digital asset class, culminating in an unprecedented maturation during 2025 and early 2026. In this scenario, the iShares Bitcoin Trust (IBIT), managed by BlackRock, has consolidated itself as the reference vehicle for institutional and retail investors seeking **regulated exposure** to the price of Bitcoin, surpassing \$70 billion in assets under management (AUM) and setting new liquidity standards for crypto derivatives. Bitcoin is no longer an 'exotic' outlier; it is a

mainstream institutional asset, thus relevant to study. This analysis aims to provide a technical assessment of IBIT call options, using the binomial model to compare three- and six-month maturities.

iShares Bitcoin Trust Fundamentals and ETF Ecosystem Dynamics

The iShares Bitcoin Trust was designed to reflect the price performance of Bitcoin, net of fees and fund liabilities, using the CME CF Bitcoin Reference Rate – New York Variant as the benchmark. One share of IBIT corresponds to approximately 0,00055BTC. Launched in January 2024, the iShares Bitcoin Trust removed the operational, tax, and custody barriers that had historically limited access to Bitcoin, allowing trading on regulated exchanges such as the Nasdaq. The ETF structure features a management fee (Sponsor Fee) of 0.25%, positioning it as one of the lowest-cost options in the U.S. market. IBIT's liquidity is a key factor in the efficiency of the options market. With daily trading volumes often exceeding 46 million shares and extremely tight bid-ask spreads (0.03% as of March 2026), the fund ensures efficient price discovery, which is reflected in the relative stability of its options compared to contracts traded on unregulated crypto exchanges. However, it's crucial to note that IBIT does not trade 24 hours a day, unlike the underlying Bitcoin, introducing gap risk during U.S. market closing hours (particularly visible after the 47% drawdown of 2026).

fund characteristics	Values in March 2026	Details
Ticker	IBIT	Traded on the Nasdaq Exchange
Benchmark	CME CF Bitcoin Reference Rate	New York Variant (BRRNY)
Management Fee	0,25%	sponsor fee
Asset Under Management	70 Billion	Allows for a very liquid market
Avg. Daily Volume (30d)	~75.5 million shares	
Avg. Bid/Ask spread	0,03%	extremely narrow spread
Tracking Error (YTD)	-19,19%	Measured relative to the Bitcoin Spot

Evaluation Method: binomial model - one time interval

To evaluate the price of the call options we used the classical binomial model. For the Historical prices of IBIT we used the data from [Yahoo Finance](#), while for the interest rates (Treasury bill) [Global Rates](#) (CME Term SOFR 3/6 Months). We considered simple compounding. We needed to compute the following quantities:

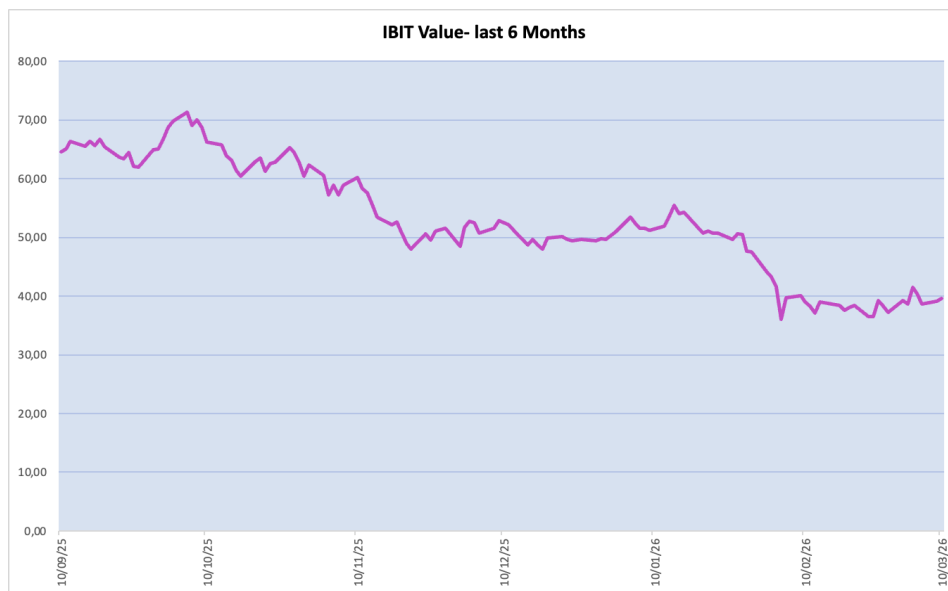
1. **Up Factor (u):** $u = e^{\sigma\sqrt{\Delta t}}$
2. **Down Factor (d):** $d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}$
3. **Risk Neutral Probability (q):** $q = \frac{(1 + R\Delta t) - d}{u - d}$

where:

- σ is the (approximated) annual volatility of the underlying. We first approximated the daily volatility, by computing the standard deviations of the daily returns $((S_i - S_{i-1})/S_{i-1})$ and then computed the annual one multiplying by $\text{sqrt}(252)$.
- r is the risk neutral interest rate
- Δt is the duration of the single time step expressed in years, which in our case were 3/12 and 6/12

The value of the call option today is derived by discounting the expected value of future payoffs at expiration:

$$C = (1 + R\Delta t)^{-1}[qC_u + (1 - q)C_d]$$



Call Option Pricing - 3 Months

Underlying Value S_0	39,71 USD	Closure of the 10/03/26
volatility σ_y	0,585	here I used 61 data to approximate σ_{daily}
Strike price K	40 USD	At the money
r	3,66%	Treasury Bill 3 months
Maturity T	3/12 Y	3 months
u	1,340	indicates a 34,02% increase
d	0,746	indicates a 25,39% decrease
q	0,442	
TARGET PRICE	4,475 USD	Bid:4,44 Ask:4,55 implied volatility: 55,42%
COMPUTED PRICE	5,800 USD	

Analysis

The theoretical target price of \$5.80 exceeded the \$4.44 market price primarily due to using high historical volatility (58.57%). **The market utilized a lower implied volatility (55.42%)**, signaling an **expected normalization** of the early 2026 turbulence. Furthermore, a single-step binomial model lacks the necessary granularity to model Bitcoin's complex short-term price distributions. Institutional "fear premiums" and the volatility skew also shifted market prices away from purely historical mathematical valuations.

Call Option Pricing - 6 Months

Underlying Value S_0	40,37	Closure of the 14/03/26
volatility σ_y	0,517	here I used 125 data to approximate σ_{daily}
Strike price K	45 USD	Out of the money*
r	3,64%	Treasury Bill 6 months
Maturity T	6/12 Y	6 months
u	1,441	indicates a 44,19% increase
d	0,693	indicates a 30,65% decrease
q	0,433	
TARGET PRICE	5,15 USD	Bid:5,05 Ask:5,25 implied volatility: 55,66%
COMPUTED PRICE	5,62 USD	

Analysis

The 6-month theoretical price (\$5,62) aligns much closer to the market premium (\$5.15) than the 3-month contract, demonstrating that extending the historical volatility window (51.7% over 125 days) improves model accuracy. However, unlike the 3-month scenario where short-term panic skewed valuations, the 6-month market data reveals a forward-looking "fear premium" with an elevated implied volatility of 55.66%. This indicates that while traders expect the immediate early-2026 turbulence to cool, they foresee sustained macroeconomic uncertainty over the next half-year. Finally we notice that even though the implied volatility is higher than the historical volatility entered into the model, the market price of the option is lower than the theoretical price, which is mathematically incorrect. The discrepancy is due to the dominant impact of other market variables.

*I used this strike because Yahoo finance considered 40 as an In the Money option, and the difference from the target price and the computed one was larger.